

Kathryn Herr, Ph.D.

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**I have right to work in the UK through the High Potential Individual visa,
which I have been granted.**

Professional Summary:

Research skills in host-microbe symbiosis, evolutionary biology, organismal ecology, microbiology, entomology, and bioinformatics. Independently written funded proposals for research, published several research articles, and co-developed the Write A Researcher pen pal outreach program. **Mentoring skills** from experience working with undergraduate researchers from disadvantaged backgrounds for four years. **Teaching skills** from instructing both introductory and advanced biology courses as well as bioinformatics workshops and volunteer groups. **Passionate** about making science and learning more efficient, reproducible, and accessible to others.

Research Experience:

Oct. 2024 – Jan. 2025	IRACDA FIRST Postdoctoral Teaching Fellow	Emory University , Dept. of Biology, <i>Atlanta, GA, USA</i>
Aug. 2018 – Aug. 2024	Ph.D. <i>Dissertation:</i> The symphony of symbiosis: Ecological and evolutionary impacts of microbial interactions with insects.	Cornell University , Dept. of Microbiology, <i>Ithaca, NY, USA</i>
Aug. 2012 – May 2016	Bachelor of Arts Major in Biology with minors in Chemistry & Chinese <i>Thesis:</i> Secreted extracellular polysaccharide layer prevents phage binding on <i>Caulobacter crescentus</i> .	Willamette University , Dept. of Biology, <i>Salem, OR, USA</i>

Selected Publications:

ORCID: 0000-0001-9647-5104

- Herr K**, Schieber J, Hendry T.A. (2024). Common fluorescent *Pseudomonas* in the phyllosphere can influence aphid behavior in diverse ways. *bioRxiv*, 2024.04.26.591271.
<https://doi.org/10.1101/2024.04.26.591271>
- Smee, M. R., **Herr, K. L.**, Abdul-Khaliq, S., Cadet, C. E. A., & Hendry, T. A. (2024). Draft genome sequences of eight bacterial isolates from *Pisum sativum* leaf surfaces. *Microbiology resource announcements*, 13(1), e0081323. <https://doi.org/10.1128/mra.00813-23>
- Moeller, A. H., Ivey, K., Cornwall, M. B., **Herr, K.**, Rede, J., Taylor, E. N., & Gunderson, A. R. (2020). The lizard gut microbiome changes with temperature and is associated with heat tolerance. *Applied and Environmental Microbiology*, 86(17). <https://doi.org/10.1128/AEM.01181-20>
- Herr, K. L.**, Carey, A. M., Heckman, T. I., Chávez, J. L., Johnson, C. N., Harvey, E., Gamroth, W. A., Wulfig, B. S., Van Kessel, R. A., & Marks, M. E. (2018). Exopolysaccharide production in *Caulobacter crescentus*: A resource allocation trade-off between protection and proliferation. *PLOS ONE*, 13(1), e0190371. <https://doi.org/10.1371/journal.pone.0190371>

Professional Experience:

Production Associate I

July 2017 - July 2018
(1 year)

Blue Heron Biotech, LLC.
Bothell, WA, USA

Worked within a multi-disciplinary team to perform DNA synthesis and analysis of construct sequences; met goals for accuracy, production deadlines and throughput utilizing the company's proprietary gene-synthesis platform and related tools. Performed Sanger sequencing and assisted in maintenance and upkeep of pipetting robots.

Field Technician

March 2017 – April 2017
(1 month)

Willamette University,
Department of Biology
Salem, OR, USA

Collaborated as part of a five-person team living in close conditions with limited water supplies and extreme weather conditions to research the coevolution of Joshua trees (*Yucca brevifolia*) and yucca moths (genus *Tegeticula*) in Delamar Valley, NV. Isolated branches of *Y. brevifolia* and took daily surveys of insect populations. Made transects and counted *Y. brevifolia* seedlings across various ecological gradients. Collected samples of *Tegeticula* and *Y. brevifolia* buds for transport back to lab. Cooked a dinner for the crew and did not make anyone sick.

Laboratory Technician

June 2016 – January 2017
(8 months)

Willamette University,
Department of Biology
Salem, OR, USA

Performed general lab management and organizational duties while researching evolutionary competition in *Caulobacter crescentus*. Utilized confocal microscopy, fluorescent proteins, Tecan plate readers, and ImageJ scripts to investigate the depths of extracellular polysaccharide layers for diverse *C. crescentus* strains. Performed plaque assays to assess the infectivity of bacteriophage to these diverse strains. Acted as supervisor for three junior research students and advised their independent research projects. Wrote and published research as a first author publication.

Teaching Experience:

Instructor – Morehouse College

January, 2024 &
August 2025

BIO306: Principles of Microbiology

- Redesigned and co-taught this upper-level biology course during both Fall and Spring semesters.
- Incorporated active learning strategies and student-centered teaching approaches to this previously lecture-based course.
- Redistributed evaluations points across multiple small assignments to make course more accessible for students.
- Created, marked, and developed rubrics for seven independent assignments, one group project, and four unit exams.
- Individually taught Unit 4: Microbial Interactions and Applied Microbiology (10 class periods)
- Co-taught Unit 1: Microbial History, Cell Structure and Function (10 lectures)
- Held office hours twice a week (1hr in person, 1hr virtual) and met with students one-on-one.

Guest Lecturer – Cornell University

September, 2023

Computational Approaches for Microbial Systems BIOMI 6300

- Designed learning outcomes and developed a case study from my own research to teach upper-level approaches to bioinformatic analysis and interpretation of next generation sequencing data.

- Utilized active learning strategies (think-pair-share, popcorn, PollEverywhere survey) to engage students, promote in-class discussion, and assess student take-aways.
- Learning outcomes:
 - Learn how both genomic and metagenomic sequencing can be harnessed to explore hypotheses.
 - Explore trade-offs from different types of sequencing (genomic vs. metagenomic) and different levels of sequencing (shallow vs. deep).

Workshop Helper – Cornell University

July, 2023

Software Carpentry Workshop (Summer)

- Provided one-on-one assistance to workshop attendees throughout the 3-day course.
- Aided learners in navigating the download, installation, and basic use of R, R Studio, R Cloud, and Github.
- Troubleshot various issues with software installation across Mac, Windows, and Linux OS, as well as issues experienced with Git commits and repository building.

Guest Lecturer – Cornell University

August, 2022 &
May, 2023

Investigative Biology BIOG 1500 (Lecture)

- Provided guest lecture to the full course (300+ students) twice a semester.
- Utilized my own research and engagement strategies to walk students through the scientific method and how to build an appropriate hypothesis.
- Asked students to identify and discuss the hypothesis, predictions, independent and dependent variables, and techniques used in my work to connect the concepts we learned in lab with my lecture.

Graduate Teaching Assistant – Cornell University

August, 2022 –
May, 2023

Investigative Biology BIOG 1500 (Lab)

- Instructed two 3-hour, 18-student lab sections a week in introductory biological concepts and techniques.
- Employed various active learning strategies to engage students and developed my own pedagogical techniques.
- Guided students through small group research projects in which they wrote and designed research proposals around green algae and nutrient limitation and then conducted independent research projects.
- Instructed students in critical thinking skills, the scientific method, how to find and read primary research articles, experimental design, writing scientific proposals, conducting independent research, sterile technique, microscopy, data collection, data analysis (R, Excel, Google Sheets), making results figures, writing scientific articles, the scientific review process, science communication (poster design and presentation, PowerPoint presentation), and career development (how to email professors, identify transferrable skills, and apply for research positions)
- Graded and provided constructive feedback on written research proposals, assignments, exams, and projects.
- Worked with and mentored Undergraduate Teaching Assistants who aided in class instruction.

Graduate Teaching Assistant – Cornell University

January, 2021-
May, 2021

Microbiology of Human Contagious Diseases BIOMI 2600 (Remote/Online Instruction)

- Facilitated discussions about topics in infectious disease and microbiology.
- Provided one-on-one instruction through office hours and small group review sessions.

- Developed exam questions and graded assignments to assess student learning.
- Transferred digital lecture and exam materials to the Canvas platform for remote instruction and assessment.
- Used Canvas tools to analyze student learning and identify areas of weakness to improve instruction.

Graduate Teaching Assistant – Cornell University

August, 2019 –
May, 2020

College Dorm Microbiology BIOMI 1100 (Partially Remote/Online)

- Lead three 2-hour weekly student discussions of case studies and concepts in microbiology that relate to the world around them.
- Guided students through writing research proposals and designing experiments to sample and culture a microbial community of their choice (with limitations to prevent the culture of pathogens). Students conducted independent research projects based on these proposals, collected their samples (swabs) and received 16S data generated from their samples.
- Instructed students in analysis and write-up of their experimental data into scientific articles.
- Graded and provided constructive feedback on research proposals and assignments.

Guest Lecturer – Cornell University

June, 2019, 2020,
& 2021

Microbial Friends and Foes Summer Research Experience for Undergrads (REU) – How to read scientific articles

- Designed and implemented my own workshop to teach students from underrepresented backgrounds how to read and understand scientific research articles.
- Assigned students a scientific article to read prior to lecture and had them identify key terms to bring to class.
- Provided guidance in how to approach research articles, and instruction in how to create concept maps, then split the class into small groups and had them create concept maps of the research article on poster paper and sticky notes.

Graduate Teaching Assistant – Cornell University

May, 2019 –
July, 2019

Microbial Friends and Foes Summer REU

- Aided the instruction of undergraduates from underrepresented and minority backgrounds in scientific research and basic concepts in microbial symbiosis.
- Served as the liaison between students and the course instructors and held office hours to provide one-on-one support to students.
- Graded assignments and progress reports.

Graduate Teaching Assistant – Cornell University

August, 2018 –
January, 2018

Public Health Microbiology BIOMI 2500

- Assisted in instruction in a flipped classroom, where students engage in problem-based learning during class and are introduced to concepts through coursework.
- Guided students through case studies in class, nudging them towards correct lines of reasoning when off track.
- Encouraged students to think critically and provided guidance in how to learn new topics without guided instruction.

Teaching Assistant – Willamette University

August, 2015 –
May, 2016

Microbiology BIO250 (Lab)

- Provided support and guidance to students in developing and conducting independent experiments in microbiology.
- Set up the laboratory space in preparation for class and made all medias and reagents.
- Graded homework assignments and provided feedback.

Teaching Assistant – Willamette University

January, 2015 –
May, 2015

Cell Biology & Genetics BIO130

- Held office hours to support students and answer questions and conducted exam review sessions twice a semester.
- Graded homework assignments and quizzes.
- Created illustrations depicting biology puns on the whiteboard each week.

Professional Development in Teaching:

Name	Host	Subject	Role	Year
Project Management for Post Doctoral Researchers	Emory University	Project Management, Mentoring	Student	Aug. 2025 – Dec. 2025
FIRST How To Teach	Emory University	Teaching, Pedagogy	Student	Jan. 2025 – May 2025
BIOG6500: Biology Instructor Training	Cornell University	Teaching, Pedagogy	Student	Aug. 2022 – Dec. 2023
Teaching & Learning in the Diverse Classroom	Cornell University	Teaching, DEI	Student	Sep. 2022 – Nov. 2022
ENTOM4040: Diversity, Equity, & Inclusion	Cornell University	Teaching, DEI	Student	Aug. 2020 – Dec. 2020

Mentorship Experience:

Overall, I have advised and supported seven undergraduate researchers in their independent projects during my scientific career. Most recently, I served as a mentor for an honors undergraduate research student during my postdoctoral studies at Emory. Previously, I mentored three undergraduate researchers across six years in the Hendry Lab, as well as three summer undergraduate researchers. Mentoring these students is a huge source of joy, and mentoring informs my teaching pedagogy. I am deeply committed to supporting and encouraging diverse students in pursuing both research careers and careers outside of academia.

Outreach Activities:

Write a Researcher (WAR)	Vice President	2021 - 2023
<i>Penpal program that connects high school students across the country with research mentors at Cornell University.</i>		

Worked with Caroline Steingaurd to help her develop and implement this program in 2021, with the goal of increasing diverse representation in STEM. We negotiated with Cornell to waive the graduate school application fee for all our high school participants. In the first year of operations, WAR supported 200 high school participants from across the country.

Graduate Student School Outreach Program (GRASSHOPR) <i>An outreach program to connect researchers in STEM fields with K-12 students from underrepresented districts in Tompkins County, NY</i>	Teaching Fellow	2019 - 2023
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In 2019 and 2021, worked as part of a three-person team to develop a three-day lesson plan to teach a 4th grade class at Beverly J. Martin Elementary about decomposition and composting, germs, and immunity, and how diet influences health. In 2022, led a team of six other fellows in developing and teaching 3rd grade students at Interlaken Elementary. Our lesson covered the history of Microbiology, in which we focused on diverse and underrepresented microbiologists.

Expanding Your Horizons (EYH) <i>EYH is a yearly conference for up to 500 students with backgrounds underrepresented in STEM with the goal of increasing representation in the STEM fields.</i>	Buddy	2019 - 2024
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Escorted 7th –9th grade students with an immersive day of scientific learning and expose them to STEM subjects not covered in their school curriculums.

Awards and Fellowships:

Awards and Fellowships:	Amount & Institution
<u>NIH IRACDA-FIRST Postdoctoral Fellowship</u> NIH Institutional Research and Academic Career Development Award (IRACDA) Fellowships in Research and Science Teaching (FIRST)	Four years of guaranteed funding provided through the FIRST program. Emory University
<u>Provost Diversity Fellowship for Advanced Doctoral Students (2023)</u> A competitive one-term fellowship awarded to advanced Ph.D. students who have made significant contributions to Cornell's core value to provide a community of inclusion, belonging, and respect where scholars representing diverse backgrounds, perspectives, abilities, and experiences can learn and work productively and positively together.	\$16,247.00 Cornell University (Stipend and tuition for the Spring 2024 semester)
<u>Graduate Student Research Enhancement Award (GSREA) – Enhancement Award (2021)</u> <i>Title: Examining the diversity of burying beetles (<i>Nicrophorus</i> spp.) and their associated <i>Yarrowia</i> symbionts in the Adirondack Mountains</i>	\$1,991.00 The Coleopterists Society
<u>Kieckhefer Adirondack Fellowship (2020)</u> <i>Title: Examining the diversity of burying beetles (<i>Nicrophorus</i> spp.) and their associated <i>Yarrowia</i> symbionts in the Adirondack Mountains.</i>	\$979.00 Cornell University
<u>Microbiome Hackathon; Best Presentation (2020)</u>	\$100.00 Cornell University
<u>Cornell CALS Outstanding TA Award (2020)</u> Awarded as Graduate Teaching Assistant for BioM1100: College Dorm Microbiology and BioMI 2500: Public Health Microbiology.	\$100.00 Cornell University

Learned Programs & Software:

Teaching & Instruction	Canvas, BlackBoard, PollEverywhere, Google Jamboard, RStudio Cloud, Jupyter Lab, Binder, Github, Zoom
Programming	Bash, Python 3.9.6
Version Control	Github Desktop, Docker, Singularity, Jupyter
Quality Control of NGS Data	FastQC (reads), CheckM (bins), BUSCO (genomes)
Metagenomic Assembly	SPAdes, BWA, Bowtie2, SAMtools, BEDtools
Genomic Assembly	Discover, Aybss, Masucra
Gene Annotation	RAST, GeneMark, RepeatMasker, Maker, Braker2, BLAST (Basic Local Alignment Search Tool), Kaiju
Data Organization & Analysis	Microsoft Excel, Google Sheets, R Markdown, PRISM, R-studio, SYSTAT 13
Phylogenetic Analysis	Phylophaln, Metaphlan, MEGA
Data Figures and Presentation	Microsoft Office (Publisher, OneNote, PowerPoint, Word, Access), Google Drawings, Adobe Illustrator, Inkscape

Conference Presentations:

- 2024 Herr, K.L. The Symphony of Symbiosis: Ecological and Evolutionary Impacts of Microbial Interactions with Insects. Cornell Institute for Host-Microbe Interactions and Disease (CIHMID) Summer Symposium. Cornell University, Ithaca, NY.
- 2022 Herr, K.L. and T.A. Hendry. An investigation into Burying beetle (*Nicrophorus* sp.) symbionts. Evolution Meeting. Cleveland, OH

Selected Presentations:

- 2024 Herr, K.L. The Symphony of Symbiosis: Ecological and Evolutionary Impacts of Microbial Interactions with Insects. Dissertation Seminar. Cornell University, Ithaca, NY.
- 2023 Herr, K.L. Experimental Design: An investigation into Burying beetle (*Nicrophorus* sp.) symbionts. BioG1500: Investigative Biology Laboratory Lecture. Cornell University, Ithaca, NY. **(Invited Lecturer)**
- 2023 Herr, K.L. and T.A. Hendry. An investigation into *Yarrowia*-like yeast symbionts of Burying beetles (*Nicrophorus* sp.). Microbiology Department Seminar. Cornell University, Ithaca, NY.
- 2023 Herr, K.L. How to read scientific articles. Microbial Friends and Foes Summer REU Lecture. Cornell University, Ithaca, NY. **(Invited Lecturer)**
- 2023 Herr, K.L. Pseudomonads in the snap pea phyllosphere. BioMi6300: Computational Approaches for Microbial Systems. Cornell University, Ithaca, NY. **(Invited Lecturer)**

- 2021 Herr, K.L. How to read scientific articles. Microbial Friends and Foes Summer REU Lecture. Cornell University, Ithaca, NY. **(Invited Lecturer)**
- 2021 Herr, K.L. Abundance of *Pseudomonas* in the snap pea (*Pisum sativum*) phyllosphere and impact of various siderophore fluorescences on aphid avoidance. Microbiology Department Seminar. Cornell University, Ithaca, NY.
- 2020 Herr, K.L. The *pseudomonas*, the phyllosphere, and the snap pea. Microbiology Department Seminar. Cornell University, Ithaca, NY.

Posters:

Herr KL, Hendry TA. Investigating the relationship between burying beetles (*Nicrophorus* sp.) and associated microbial symbionts. Animal-Microbe Symbioses, Gordan Research Conference 2023. June 18-19th, 2023. Renaissance Tuscany Il Ciocco, Lucca, Italy.

Herr KL, Hendry TA. Evolution of Yeast Symbionts from Burying Beetles (*Nicrophorus* spp.). Animal-Microbe Symbioses, Gordan Research Seminar 2023. June 17th, 2023. Renaissance Tuscany Il Ciocco, Lucca, Italy.